

Development Operations

Assignment # 2

REG#: _____

NAME: _____

COURSE CODE: CS316

INSTRUCTOR: MUHAMMAD SAJID ALI

TOTAL MARKS: 156

DevOps Automation using Shell Scripting

Objective

The objective of this assignment is to help students apply shell scripting concepts to automate common DevOps and system administration tasks in a Linux environment.

General Instructions:

All questions are mapped to CLO2 → GA-5 → C3 (APPLY).

Important: In addition to the functionality of your scripts, you *will also be evaluated* based on the **quality of your comments**. Please provide clear, concise comments within your scripts to explain the purpose and usage of each part. Comments are crucial for understanding your code and will be considered in the marking process.

Question 1: Write the bash script named 'systeminfo.sh' that will do the following tasks. **(06 marks)**

1. Print the *kernel name, kernel release, processor type* and installed *operating system*?
2. Print the name of your favorite editor and display where that editor and its documentation is located in the filesystem.

Question 2: User and Group Management. Create a Bash script named "usergroup.sh" that performs the following actions: **(10 marks)**

1. Create a user named "devopsuser."
2. Create a group named "devopsgroup."
3. Add the "devopsuser" to the "devopsgroup."

Question 3: SSH Configuration. Create a Bash script named "sshconfig.sh" that performs the following actions: **(20 marks)**

1. Generate an SSH key pair using the RSA algorithm with the default settings. Save the private key to the created user home directory.
2. Configure SSH to disable password authentication.
3. Set up SSH so that the user can log in without specifying their username and IP address each time.
 - a. SSH into the system using the newly created credentials in non-interactive mode and do the following tasks:
 - b. Create a test directory inside the home directory.
 - c. Create a file named "filecreatedinnoninteractivemode.txt" and write "I am doing the task3."

Question 4: File and Directory Manipulation. Create a Bash script named "filedir.sh" that performs the following actions: **(10 marks)**

1. Create a directory named `"/devopsdir"`.
2. Inside `"/devopsdir,"` create a file named `"devopsfile.txt"` and write `"AoA, Hello DevOps! "` to it.
3. Give the `"devopsuser"` read and write permissions for the `"devopsfile.txt"` and read, write, and execute permissions for `"devopsdir."` Give read permissions to other users.

Question 5: Backup Script. Create a Bash script named `"backup.sh"` that will compress the contents of the `"devopsdir"` directory into a tarball (`tar.gz`) every 10 minutes and save each backup in the `"devopsuser"` home directory, specifically inside the `"backup"` directory. Each backup should have a unique and descriptive name for easy identification. **(10 marks)**

Bonus Marks: If you can write a script that not only performs the backup but also sets up the cron job for you, you will receive bonus marks."

Question 6: – System Health Report

(40 marks)

Create a Bash script that collects key system information and writes it to an HTML report.

1. Name the file `system_health.sh`.
2. Collect and display:
 - a. Current date as shown in figure
 - b. Memory usage as shown in the give figure [Hint: use `awk` utility along with other commands]
 - i. Total memory and swap memory
 - ii. Used memory and swap memory in percent
 - iii. Available memory and swap memory in percent

System Health Report		
Generated on: Wednesday, Oct 2025		
CPU Usage - Last 5 Minutes		
0.61%		
Memory Usage - Last 5 Minutes		
Total	Used	Available
7Gb	69.50%	4.03%
Swap Memory - Last 5 Minutes		
Total	Used	Available
2047Mb	50.36%	49.63%
Disk Usage - Last 5 Minutes		
Total	Used	Available
82Gb	19.01%	76.18%
Disk Partition Details - Last 5 Minutes		
Disk	Used	Available
/dev/sda8	20.28%	74.58%
/dev/sda2	31.51%	68.48%
Designed by: u2022123		

- c. Disk usage [Hint: use regex pattern in `awk` to filter output records, `!` used to negate & `^` means to start a string with chars]

- i. Total disk space
 - ii. Total disk space used and available
 - iii. List of partitioned disks available, do not include *tmpfs* disks by using `--exclude` flag
 1. Show disk name
 2. Disk space used
 3. Disk space available
3. Use heredoc to write this info into `system_health.html`.
4. Host this `system_health.html` as `index.html` using `nginx` server.
5. If successful, print success otherwise print an error and exit with code 1.

Please see the above image for the expected output:.

Question # 7: Automating the Installation of LAMP Stack Environment on Linux.

(50 Marks)

[Hint: try to use the documentation as much as possible]

1. Create a bash script file named `bsc-configuring-lamp-environment.sh` and make it executable.
2. Add Command Line Options – update this bash script file so that it should optionally:
 - a. Take **-i** as a command line argument when executing this script
 - i. `./bsc-configuring-lamp-environment.sh -i`
 - ii. This means script will install missing LAMP services.
 - b. Take **-d** as a command line argument when executing this script
 - i. `./bsc-configuring-lamp-environment.sh -d`
 - ii. This means script will delete all LAMP services from the system.
 - c. Take **-ws** with value `<Apache or Nginx>` as a command line argument when executing this script
 - i. `./bsc-configuring-lamp-environment.sh -i -ws Apache`
 - ii. This means script will install *Apache* or *Nginx* as a web service while setting up LAMP stack.
3. Web Server Installation – update this bash script file so that it should:
 - a. Install the latest version of the selected web server (Nginx or Apache) when executed without any user interaction.
 - b. By default, install Apache as a web service if `-ws` option is not provided in the command line argument.
 - c. If there's a previously installed web server, ask the user if want to remove the previous server (e.g., "Apache web server is already installed"). If the user agrees (Y or y), remove it, and proceed with the requested installation. Otherwise, skip and continue.
4. Further update the same bash script file you created earlier so that it installs the latest version of PHP when executed without any user interaction. You must take care of all the basic modules that might be required for a simple PHP based application. You must show an appropriate message to the user if it is already installed.
5. Update the same bash script file to install the MySQL database service. You must generate a root password for MySQL during the installation and display it to the user.
6. Update the same bash script file to delete all the installed LAMP stack services with a special warning about taking backups. Also, take second input from the user with a message 'are you sure, you want to continue ... (Y/n)'. If user chooses Y delete the LAMP stack otherwise skip and continue.
7. Logging Script Activities – enhance the script to maintain a log file that records all major actions performed by the script.

- a. Create a log file named lamp-setup-<your-reg>.log
- b. Record important activities in the log file, such as:
 - i. Start and end of script execution
 - ii. Installation of services
 - iii. Removal of services
 - iv. Any errors or warnings generated during execution
- c. Each log entry should include a timestamp.
 - i. [2026-03-10 14:30:21] Installing Apache Web Server

Question 8: You've created several Bash scripts as part of this assignment. Now, you need to submit all these scripts, along with this script on teams. Write a Bash script named submit.sh' that accomplishes the following: **(10 marks)**

1. Creates a tarball (compressed archive) named 'Your_Reg_No.tar.gz', which contains all the Bash scripts you've created in this assignment, including this one. Do not manually give the name of each scripted file. You may have an arbitrary number of scripts.

[Hint*: you may use patterns to match string, i.e., '*.jpg']

2. Upload the tar file on the teams

Submission

1. Use tar and gzip to create compressed bundle of all inside your home directory under name assignment2_<your_reg_number>.zip
2. Upload the tarball file on teams.